

Homework 4

Due Wednesday, 23 September 2026

1. Let $X_1, \dots, X_n$ be a random sample from the pdf

$$f(x; \theta) = \theta x^{-2}, \quad 0 < \theta \leq x < \infty.$$

- Find the method of moments estimator of θ . (hint: don't overthink this).
- Find the maximum likelihood estimator of θ .

2. Let $X_1, \dots, X_n$ be iid with pdf

$$f(x; \theta) = \frac{1}{\theta} \quad 0 \leq x \leq \theta; \theta > 0$$

Estimate θ using both the method of moments and maximum likelihood. Calculate the means and variances of the two estimators. Which one should be preferred and why?

3. Suppose that $X_1, \dots, X_n$ are a random sample from a Uniform($0, 2\theta + 1$) distribution.
- Find the MLE of θ .
 - Obtain the MLE of the **variance** of the underlying Uniform($0, 2\theta + 1$) distribution.
4. Let $Y_1, \dots, Y_n$ be iid random variables from the population with pdf

$$f(y; \beta) = \begin{cases} 3\beta^3 y^{-4} & y \geq \beta \\ 0 & \text{elsewhere} \end{cases}$$

Find the MLE of β .

5. Suppose the random variables $Y_1, \dots, Y_n$ satisfy

$$Y_i = \beta x_i + e_i, \quad i = 1, \dots, n$$

where $x_1, \dots, x_n$ are fixed constants and $e_1, \dots, e_n$ are iid $N(0, \sigma^2)$ random variables, with σ^2 unknown.

- Find the MLE of β , and find $E(\hat{\beta}_{\text{MLE}})$.
- Find the distribution (including parameters) of $\hat{\beta}_{\text{MLE}}$.